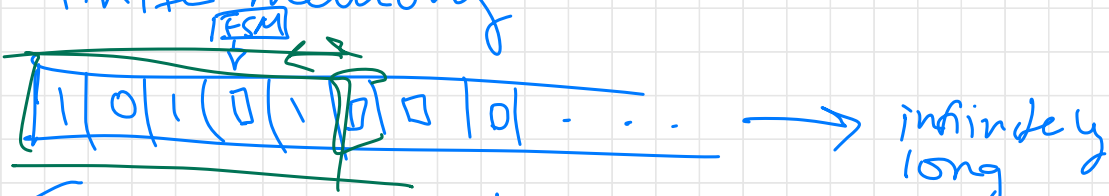
Turing Machines

finite states

infinite memory



Σ input alphabet

$\Sigma = \{0,1\}$

Γ = tape alphabet $= \Sigma \cup \{\square\} \cup \ldots$

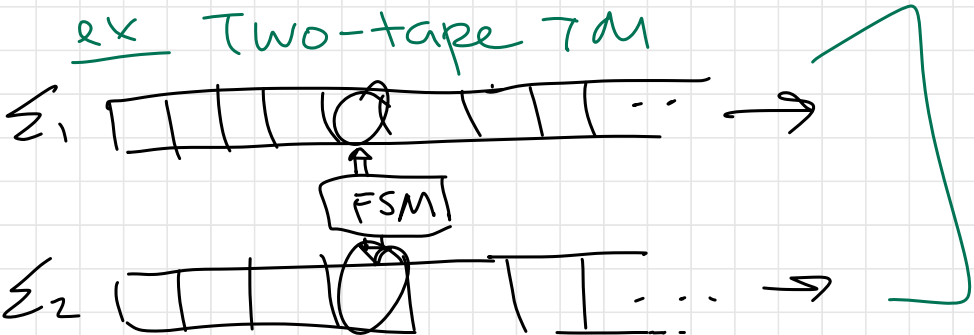
$\square$ blank symbol $\notin \Sigma$

"weaker" machines can simulate standard TMs:

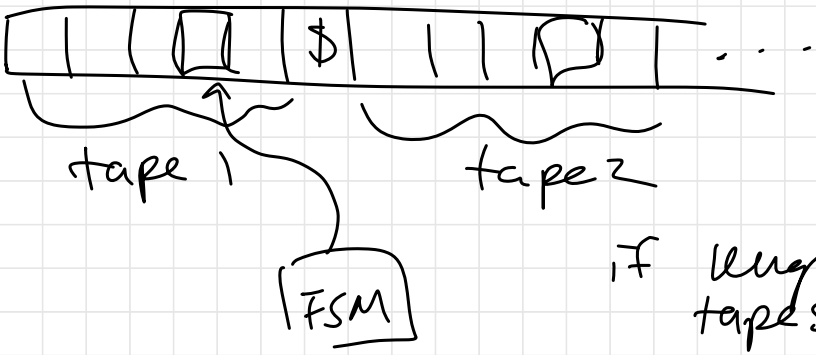
~~write~~ alter once or ~~write~~ alter twice

Standard TM can simulate "stronger" machines

ex Two-tape TM



simulate w/ single tape

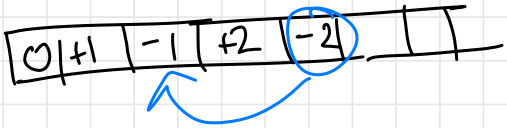


if length of tapes grows, copy tape 2 farther R

ex Doubly infinite tape

simulate w/ single tape?

Use a two-tape machine



ex

subroutines

RAM

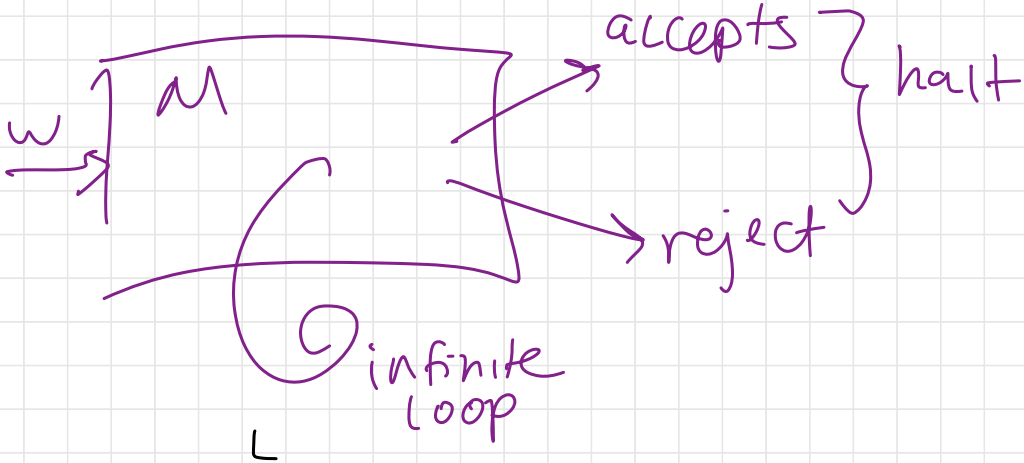
⋮

Church-Turing thesis:

all computation is equivalent to a standard TM.

Undecidability

Given a TM M:



A language L is <u>decidable</u> if there is a TM that accepts every string in L and rejects every string not in L.

A language L is <u>recognizable</u> if there is a TM that accepts a string iff it is in L.

$w \in L$

$w \notin L$

M defines 4 languages:

→ $\text{ACCEPT}(M) = \{w \in \Sigma^* : M \text{ accepts } w\}$

$\text{REJECT}(M) = \{w : M \text{ rejects } w\}$

$\text{HALT}(M) = \{w : M \text{ halts on } w\}$

$= \text{ACCEPT}(M) \cup \text{REJECT}(M)$

$\text{DIVERGE}(M) = \Sigma^* \setminus \text{HALT}(M)$

let $\langle\ \rangle$ be some encoding of TM

$\langle M \rangle$ "encoding of M"

let $\text{SELFREJECT} = \{\langle M \rangle : M \text{ rejects } \langle M \rangle\}$

can you describe:

an alg. that rejects its own encoding?

↳ pseudocode

string of Python prog.

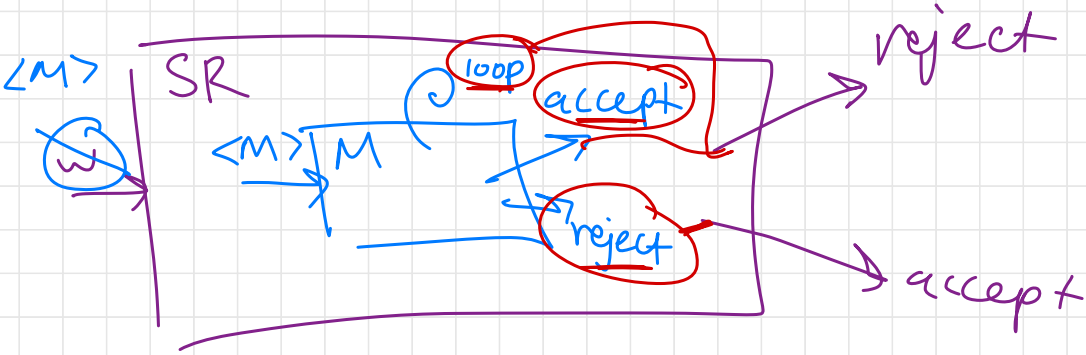
an alg that accepts its own encoding?

```
Alg(w)
  for i in 1 to 10:
      print "Hi"
  return True
```

does Alg accept itself?

```
Alg1(w):
  input = w
  if "i" in input:
      accept
  else reject
```

pure can be no deciding TM

Theorem SELFREJECT is undecidable.

Proof Suppose not. Let SR be the TM that decides SELFREJECT.